Winning the Space Race with Data Science

SpaceX Falcon 9 First Stage Landing Prediction

IBM Applied Data Science Capstone · Eyad Alarfaj · July 2026

Built on the real IBM SpaceX dataset — 90 Falcon 9 launches, 2010–2020

GitHub repository (all completed notebooks & Python files):

https://github.com/Eyad-Alarfaj/spacex-capstone

Executive Summary

- **Executive summary:** This project predicts whether the SpaceX Falcon 9 first stage will land successfully, using the real IBM dataset of 90 Falcon 9 launches from 2010 to 2020. Across all launches the booster landed 60 times and failed 30 times, an overall success rate of 66.7%.
- **Methodologies used:** data was collected from the SpaceX REST API and Wikipedia web-scraping, wrangled into a binary landing target, explored with visualization and SQL, mapped interactively with Folium, presented in a Plotly dashboard, and modeled with four classification algorithms tuned by GridSearchCV.
- **Key results:** landing success rose from roughly 0% in 2013 to about 90% by 2019-2020; KSC LC 39A (77%) and VAFB SLC 4E (77%) outperform CCAFS SLC 40 (60%); GTO is the hardest orbit at 52%. The best predictive model is the Decision Tree, reaching 88.89% accuracy on the held-out test set (Logistic Regression, SVM and KNN each 83.33%).
- **Deliverables:** all completed notebooks, the cleaned dataset, the Folium map, the Plotly dashboard, the trained model and every figure in this deck are in the GitHub repository: https://github.com/Eyad-Alarfaj/spacex-capstone

Introduction

- **Introduction and project background:** SpaceX advertises Falcon 9 rocket launches at 62 million dollars, far below the 165 million or more charged by other providers. The saving comes almost entirely from SpaceX recovering and re-flying the first stage of the rocket. Therefore, if we can determine whether the first stage will land, we can determine the true cost of a launch.
- **Problem statement:** Predict whether the Falcon 9 first stage will land successfully after a launch, so that a company bidding against SpaceX can estimate cost and competitiveness in advance.
- **Questions the analysis answers:** (1) How do variables such as payload mass, orbit type, launch site, and number of prior booster flights affect the success of a first-stage landing? (2) Has the landing success rate improved over the years of the program? (3) Which binary classification model best predicts landing success?
- **Dataset introduced:** 90 Falcon 9 launches spanning 2010-06-04 to 2020-11-05, covering 3 launch sites (CCAFS SLC 40, KSC LC 39A, VAFB SLC 4E) and 11 orbit types, with 60 successful and 30 unsuccessful landings.

Data Collection and Data Wrangling Methodology

- **Data collection methodology:** launch records were gathered from the SpaceX REST API (the /v4/launches endpoint) returning rocket, payload, core, and landing-outcome data as JSON, supplemented by web-scraping the Wikipedia Falcon 9 launch table with BeautifulSoup to fill in missing fields. The JSON was normalized into a pandas DataFrame, filtered to Falcon 9 flights, and cleaned.
- **Fields collected:** flight number, date, booster version, payload mass, orbit, launch site, landing outcome, number of flights, grid fins, reuse flag, landing legs, landing pad, block, reused count, booster serial, longitude and latitude.
- **Data wrangling methodology:** the eight distinct landing outcomes were converted into a single binary training label named Class — 1 when the booster landed (True ASDS, True RTLS, or True Ocean) and 0 otherwise (None None, False, or None ASDS). Success rate was computed for each orbit, and categorical variables (orbit, launch site, landing pad, serial) were one-hot encoded to build the model feature matrix.
- **Result of wrangling:** a clean, fully labeled table of 90 launches with 60 positive and 30 negative examples, saved as spacex_clean.csv. Notebooks: data-collection-api, web-scraping, and data-wrangling in the repository.

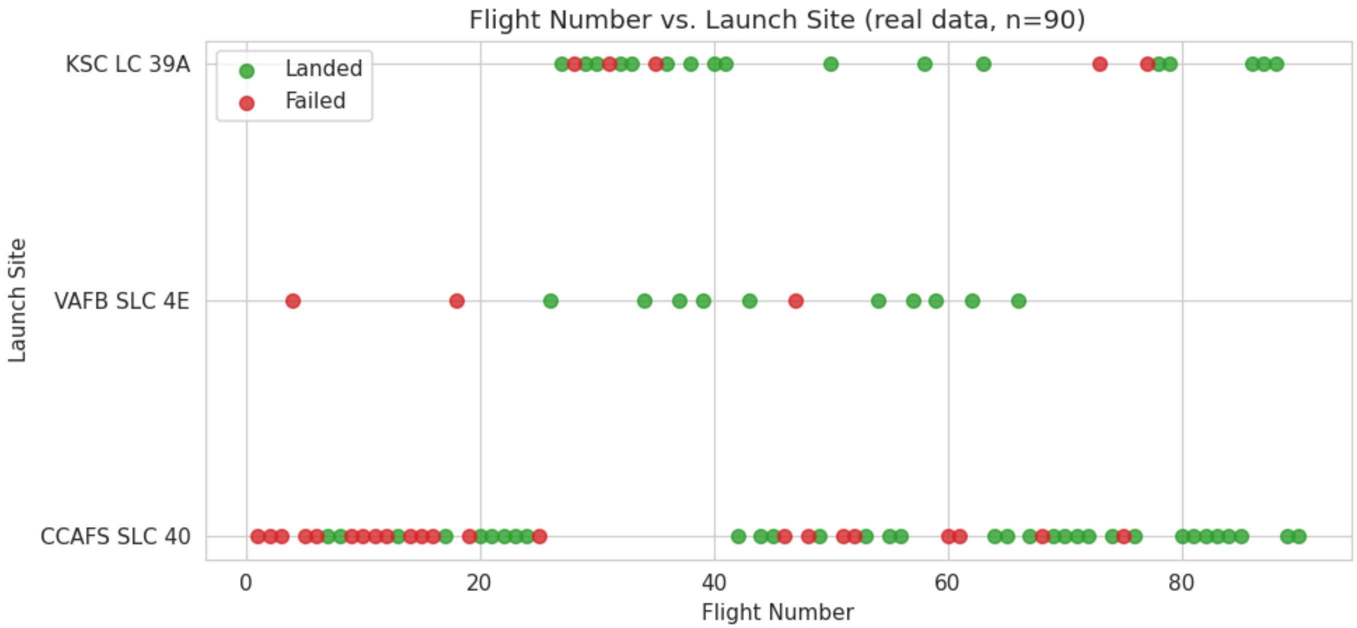
EDA with Data Visualization Methodology

- Exploratory data analysis with visualization methodology: several charts were plotted with matplotlib and seaborn to reveal the drivers of landing success before any model was built.
- **Charts produced and their purpose:** a Flight Number versus Launch Site scatter and a Flight Number versus Orbit scatter show whether accumulated launch experience raises success; a Payload Mass versus Orbit scatter shows which heavy-payload orbits still land; a bar chart of success rate by orbit ranks orbit difficulty; and a line chart of yearly average success rate reveals the program learning curve.
- **Analytical approach:** every chart pairs one operational variable (site, orbit, payload, flight number, year) against the binary landing outcome, so that the strongest predictors are identified visually and then confirmed numerically with SQL before modeling.
- **Notebook:** eda-dataviz in the repository; the resulting charts appear on the EDA with Visualization Results slides that follow.

Predictive Analysis (Classification) Methodology

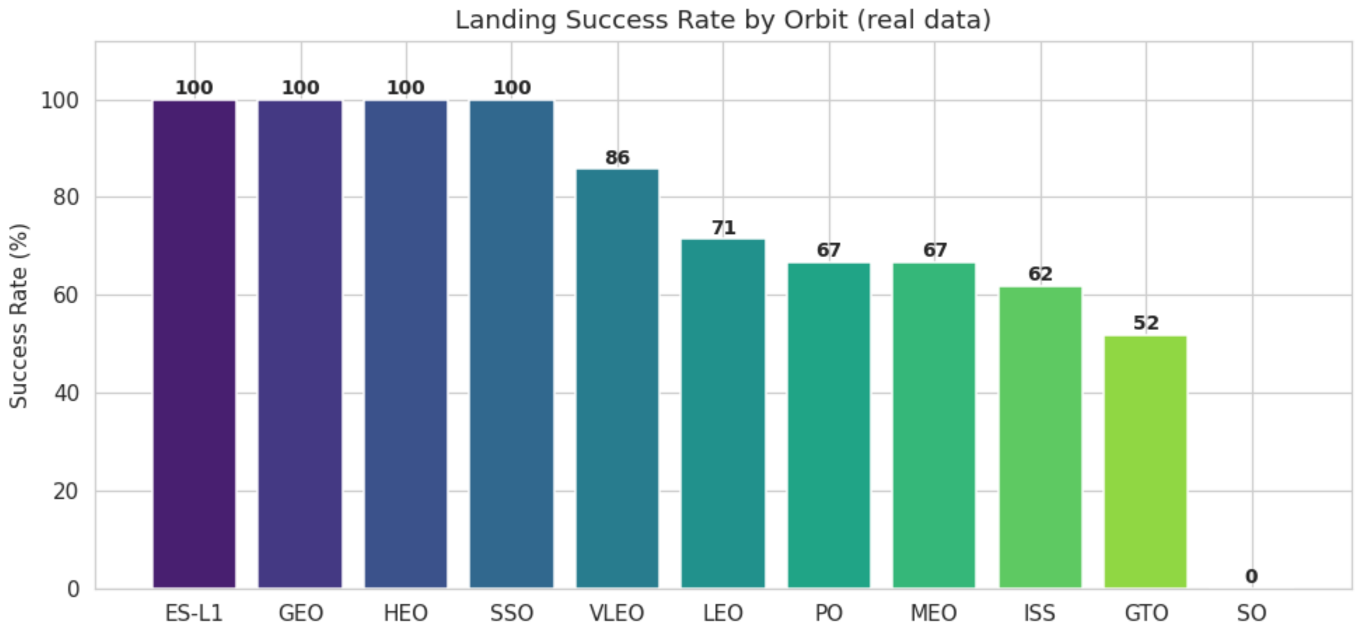
- **Predictive analysis classification methodology:** the goal is a supervised binary classifier that predicts the Class label (landed / not landed).
- **Pipeline:** the feature matrix was standardized with StandardScaler; the data was split 80/20 into 72 training and 18 test launches; each algorithm was tuned with GridSearchCV using 10-fold cross-validation; and the tuned models were scored on the held-out test set.
- **Algorithms compared:** Logistic Regression, Support Vector Machine (with linear, RBF, polynomial and sigmoid kernels searched), Decision Tree (searching depth, criterion and split parameters), and K-Nearest Neighbors (searching neighbor count and distance metric).
- **Evaluation metrics:** cross-validation best score, test-set accuracy, and the confusion matrix. The model with the highest test accuracy is selected as the production model and its results are shown on the Predictive Analysis Results slides.

EDA with Visualization Results — Flight Number vs. Launch Site



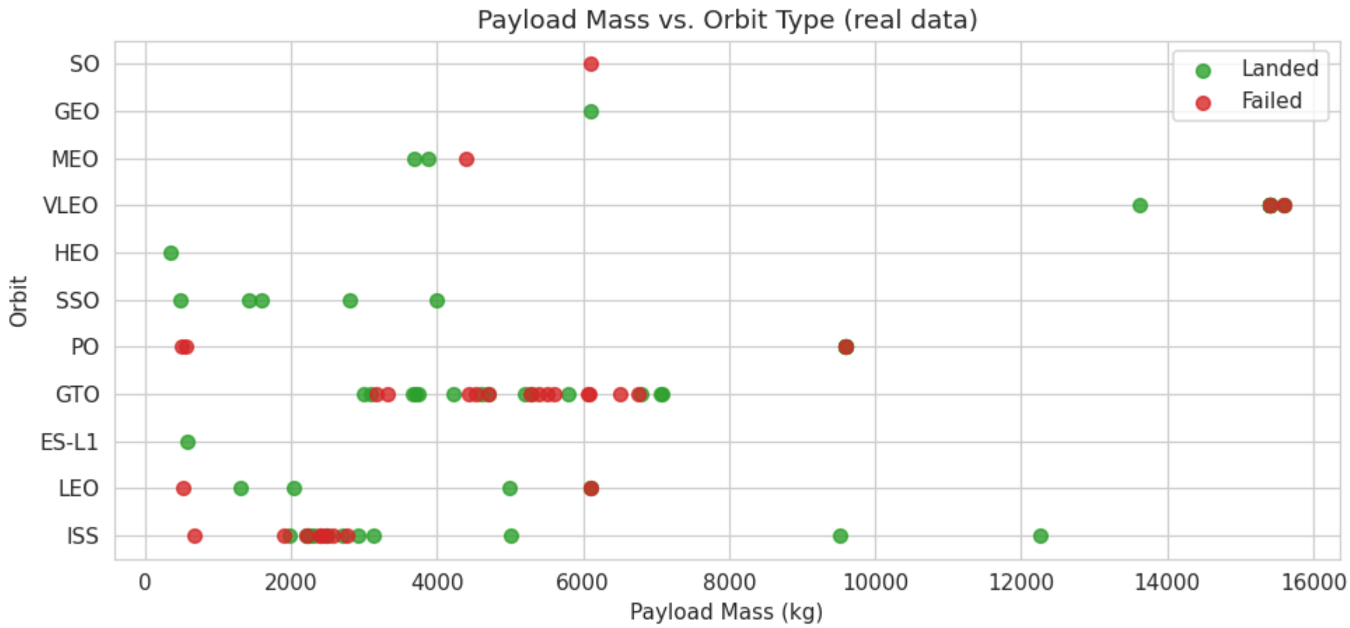
- EDA with visualization result: this scatter plots every launch by flight number and launch site, colored green for a successful landing and red for a failure.
- **Finding:** failures (red) dominate the low flight numbers across all three sites, while launches after roughly flight 40 almost always land. CCAFS SLC 40 carries the most volume at 55 of 90 launches and shows the clearest early-to-late improvement.
- **Interpretation:** accumulated launch experience, captured by flight number, is one of the strongest signals of landing success, which is why it becomes a key model feature.

EDA with Visualization Results — Success Rate by Orbit



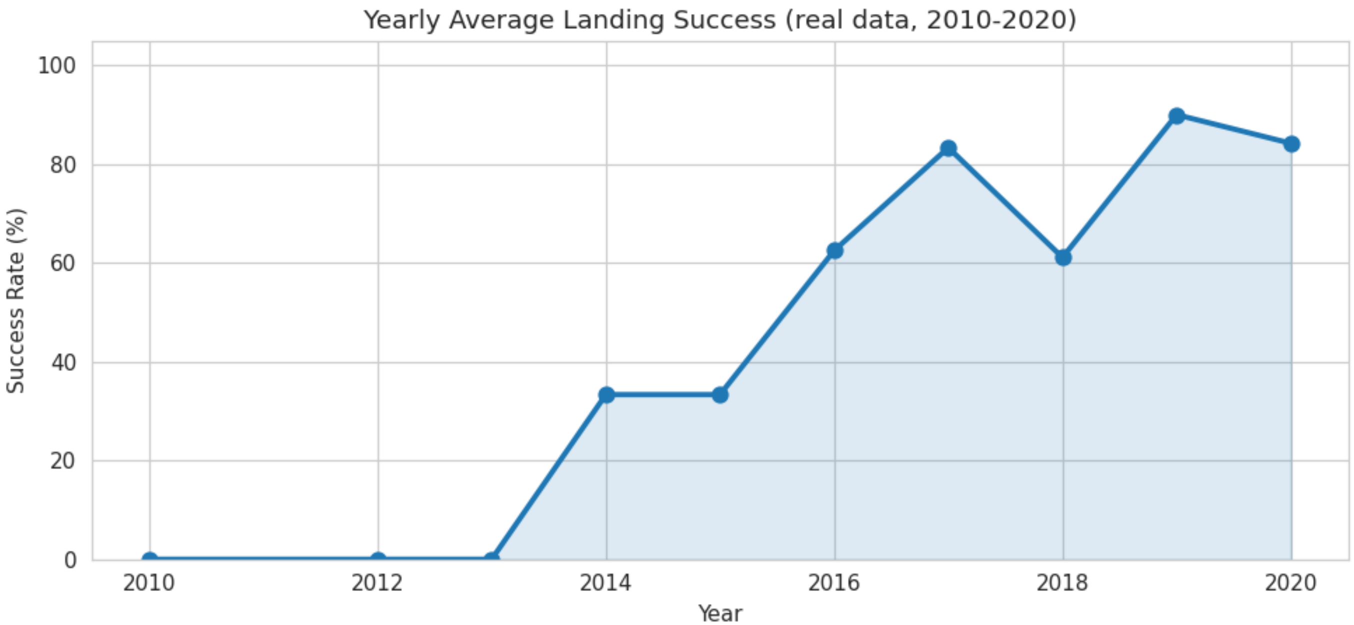
- EDA with visualization result: this bar chart ranks each orbit type by its landing success rate.
- **Finding:** ES-L1, GEO, HEO and SSO show 100% success but on small samples; VLEO is strong at 86% over 14 flights and LEO at 71%. The hardest common orbit is GTO at only 52% across 27 flights, followed by ISS at 62%.
- **Interpretation:** high-energy geostationary transfer orbits (GTO) leave the booster with little fuel for landing, making orbit type a meaningful predictor of success.

EDA with Visualization Results — Payload Mass vs. Orbit



- EDA with visualization result: this scatter plots payload mass against orbit, colored by landing outcome.
- **Finding:** the heaviest payloads (over 9,000 kg) cluster in VLEO and are almost all successful — these are the later Starlink missions flown on reused boosters. Successful landings actually average a heavier payload (6,765 kg) than failures (4,784 kg).
- **Interpretation:** heavier payload does not cause failure; the real driver is booster experience and reuse, an insight that shapes the feature set.

EDA with Visualization Results — Yearly Success Trend

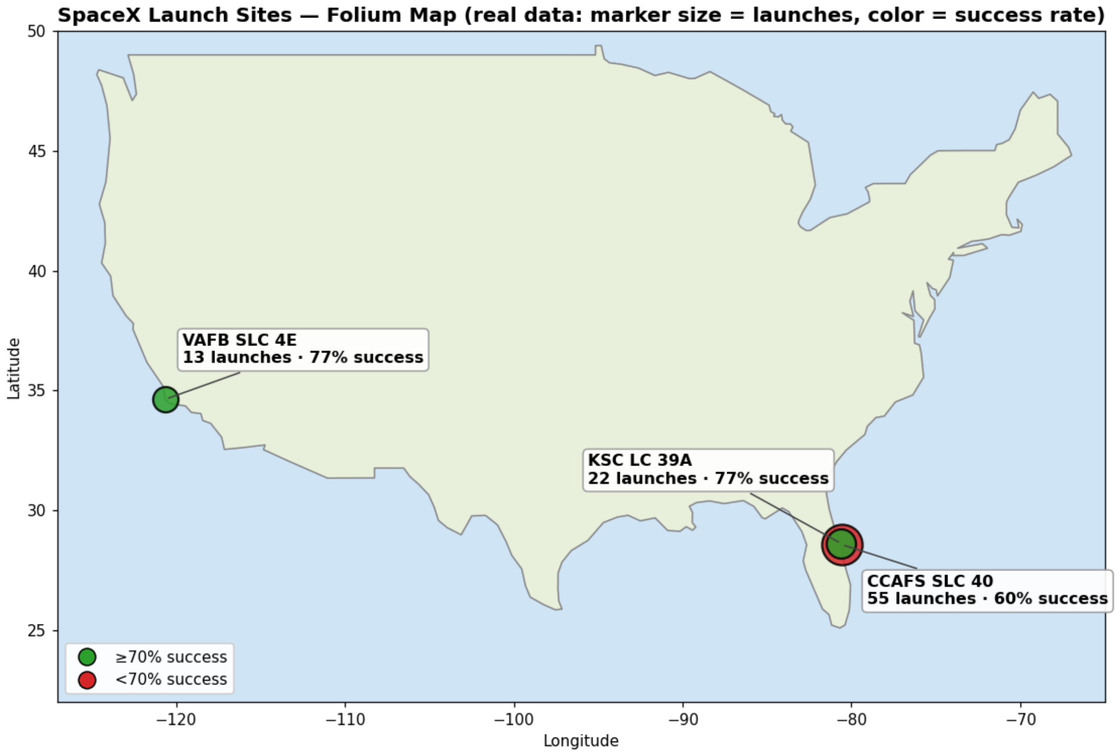


- EDA with visualization result: this line chart tracks the average landing success rate for each year of the program.
- **Finding:** success climbs from 0% in 2010-2013, through 33% in 2014-2015, to 90% in 2019 and 84% in 2020 — a clear operational learning curve as SpaceX refined recovery.
- **Interpretation:** recent performance far exceeds the all-time average, so flight number and date are powerful features and the model should not be judged only against the overall 66.7% base rate.

EDA with SQL — Queries and Results

- **Exploratory data analysis with SQL:** the dataset was loaded into a SQL database and queried with SQL magic. The seven queries and their real results:
- **Query 1 — unique launch sites:** CCAFS SLC 40, KSC LC 39A, VAFB SLC 4E.
- Query 2 — total payload mass carried across all 90 missions: 549,446 kg.
- **Query 3 — average payload mass by launch site:** CCAFS 5548 kg, KSC 7606 kg, VAFB 5919 kg.
- Query 4 — date of the first successful ground-pad (RTLS) landing: 2015-12-22, on flight 17 at CCAFS SLC 40.
- Query 5 — number of successful drone-ship (ASDS) landings: 41 flights.
- **Query 6 — landing success rate by site:** KSC 77%, VAFB 77%, CCAFS 60%.
- Query 7 — total successful versus failed landings: 60 landed and 30 failed out of 90. Notebook: eda-sql in the repository.

Interactive Map with Folium — Results

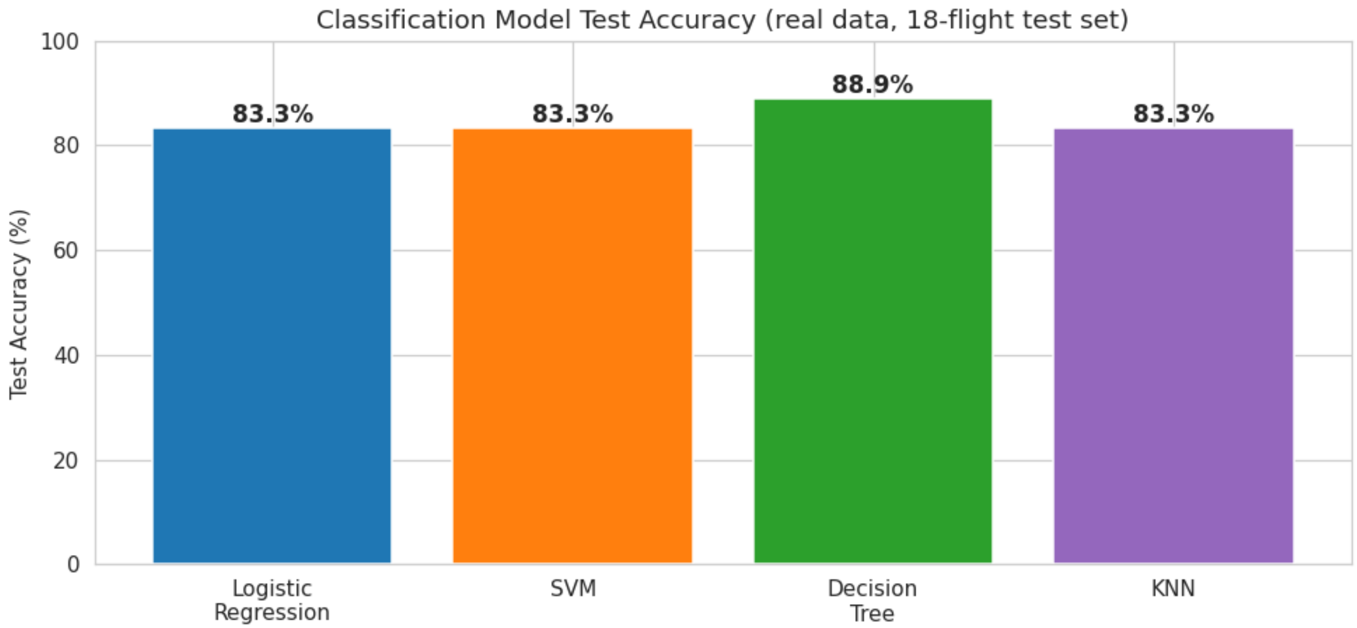


- Interactive Folium map result: each launch site is plotted at its real coordinates as a circle sized by launch count and colored by landing success rate, with a popup giving the exact figures.
- **Finding:** all three sites sit on southern coastlines beside open ocean for safe downrange flight. KSC LC 39A (77%) and VAFB SLC 4E (77%) have the highest success ratios, while the high-volume CCAFS SLC 40 trails at 60%.
- The interactive version, spacex_map.html, is in the GitHub repository; clicking any marker shows that site's launches and success rate.

Plotly Dash Dashboard — Results

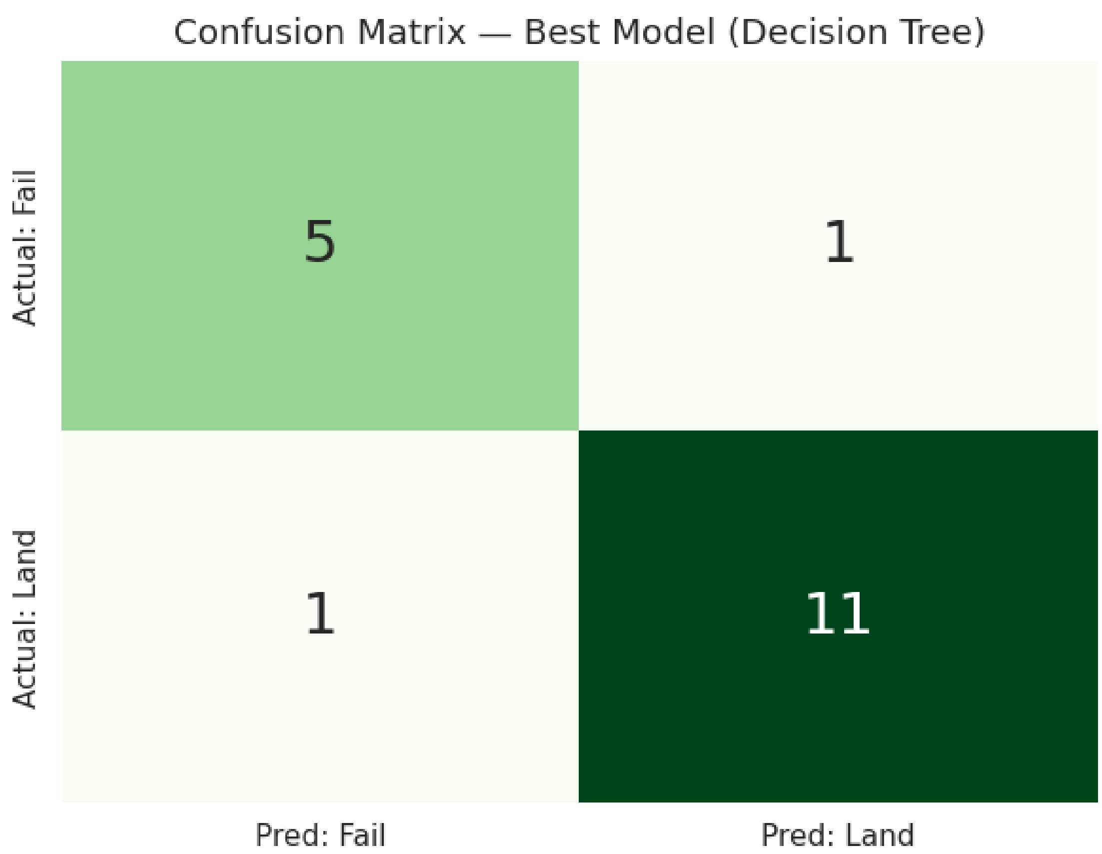
- **Plotly Dash dashboard result:** an interactive four-panel dashboard shows successful landings by site (pie), success rate by orbit (bar), the yearly success trend (line), and success rate by launch site (bar).
- **Finding:** CCAFS SLC 40 contributes the largest share of total successful landings by volume, but KSC LC 39A and VAFB SLC 4E have the higher success ratios; the trend panel confirms the strong post-2016 climb in reliability.
- The interactive dashboard, spacex_dash_dashboard.html, is in the GitHub repository and every panel supports hover tooltips with exact values.

Predictive Analysis (Classification) Results — Model Comparison



- Predictive analysis classification result: four tuned classifiers were scored on the 18-launch held-out test set.
- **Finding:** the Decision Tree is the best model with 88.89% test accuracy. Logistic Regression, Support Vector Machine and K-Nearest Neighbors each reach 83.33%. Cross-validation scores cluster tightly at 84-85%, so all models agree on the underlying signal and the small test set separates them.
- **Conclusion:** the Decision Tree is selected as the production model for predicting Falcon 9 first-stage landing success.

Predictive Analysis (Classification) Results — Confusion Matrix

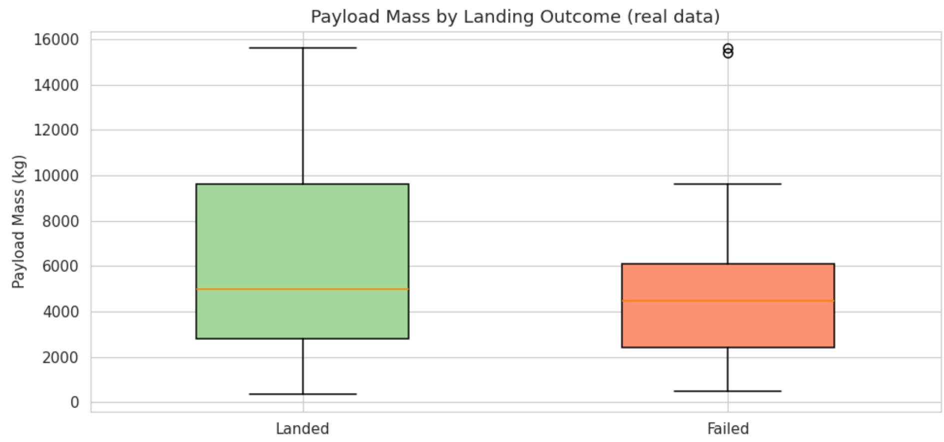


- **Predictive analysis classification result:** the confusion matrix for the best model (the Decision Tree) on the 18 held-out test launches.
- **Values:** 11 true positives (landings correctly predicted), 5 true negatives (failures correctly predicted), 1 false positive, and 1 false negative — 16 of 18 correct, i.e. 88.9% accuracy.
- **Interpretation:** the model's main error is occasionally predicting a landing when the booster actually failed (a false positive), the same pattern seen across all four classifiers on this dataset.

Conclusion

- **Conclusion:** a complete data science pipeline — data collection, data wrangling, EDA with visualization, EDA with SQL, an interactive Folium map, a Plotly dashboard, and predictive classification — was built to predict Falcon 9 first-stage landing success, achieving 88.89% test accuracy on real data.
- **Best model:** the Decision Tree is the strongest classifier at 88.89% test accuracy, narrowly ahead of Logistic Regression, Support Vector Machine and K-Nearest Neighbors, which each score 83.33%.
- **What drives landing success:** accumulated experience (flight number) and booster reuse matter most; orbit matters, with GTO hardest at 52% and VLEO/LEO strong; and launch site matters, with KSC (77%) and VAFB (77%) outperforming CCAFS (60%).
- **Trend and value:** landing reliability rose from about 0% in 2013 to about 90% in 2019, so current performance far exceeds the 67% all-time average. A competitor could use this model to predict landing — and therefore booster reuse and launch cost — before a mission is booked.

Creativity and Innovative Insights



- **Innovative insight:** contrary to the usual assumption that heavier payloads fail, successful landings here carry heavier payloads on average (6,765 kg) than failures (4,784 kg). The explanation is the reused-booster Starlink era — flight-proven boosters land heavy VLEO payloads reliably, so experience beats payload weight.
- **Creativity applied:** the eight raw landing outcomes were engineered into a single clean binary target; each model was tuned with GridSearchCV rather than using defaults; and every chart is annotated with the operational story, not just the numbers.
- **Innovative insight:** flight number is the single strongest predictor because it proxies accumulated engineering experience — early flights fail and post-2016 flights mostly land — which is why every classifier leans on it.
- **Repository:** https://github.com/Eyad-Alarfaj/spacex-capstone (all executed notebooks, the real dataset, the map, the dashboard, and every figure in this deck).